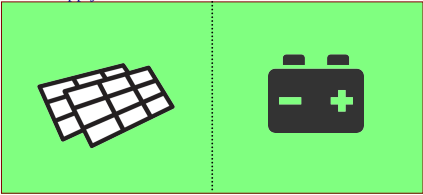
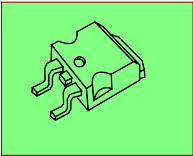


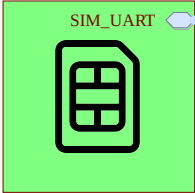
Solar and Battery Charger and Supply
Power Supply



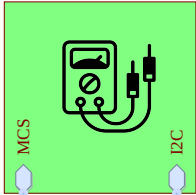
Solar to Sim module
4V2 Buck Convertor



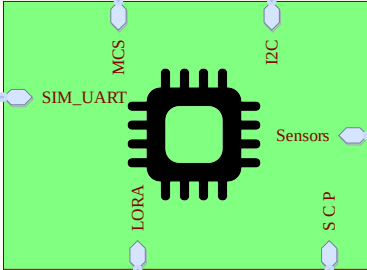
Sim Module
Sim A7670C



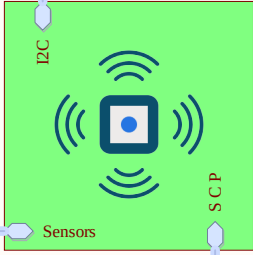
Watt meter
INA3221



ESP 32
Micro controller



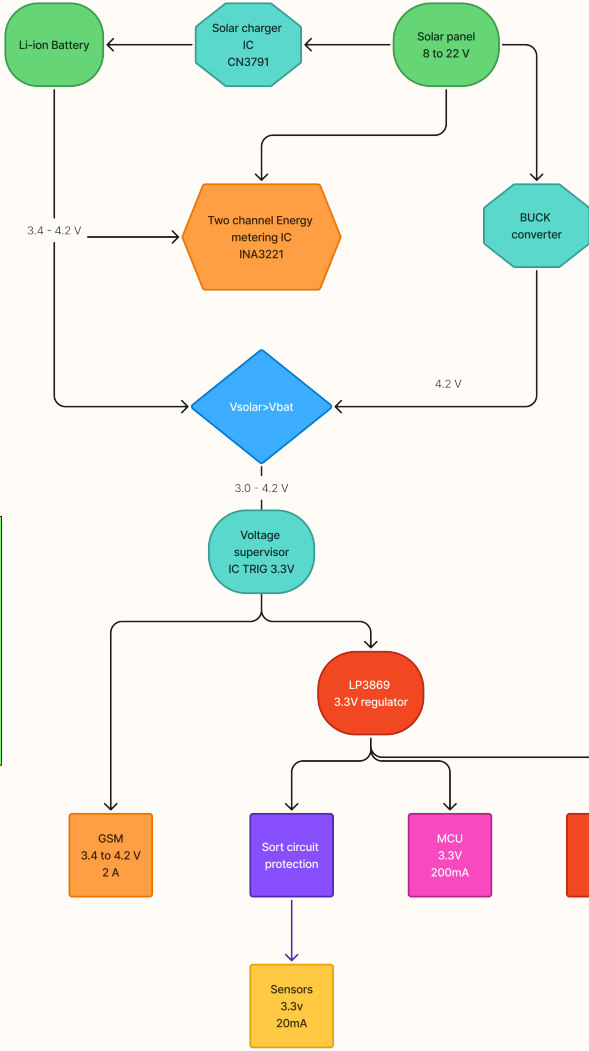
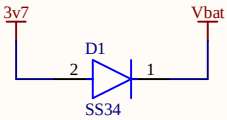
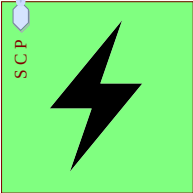
Sensors
Sensors



Lora RA-02
Lora



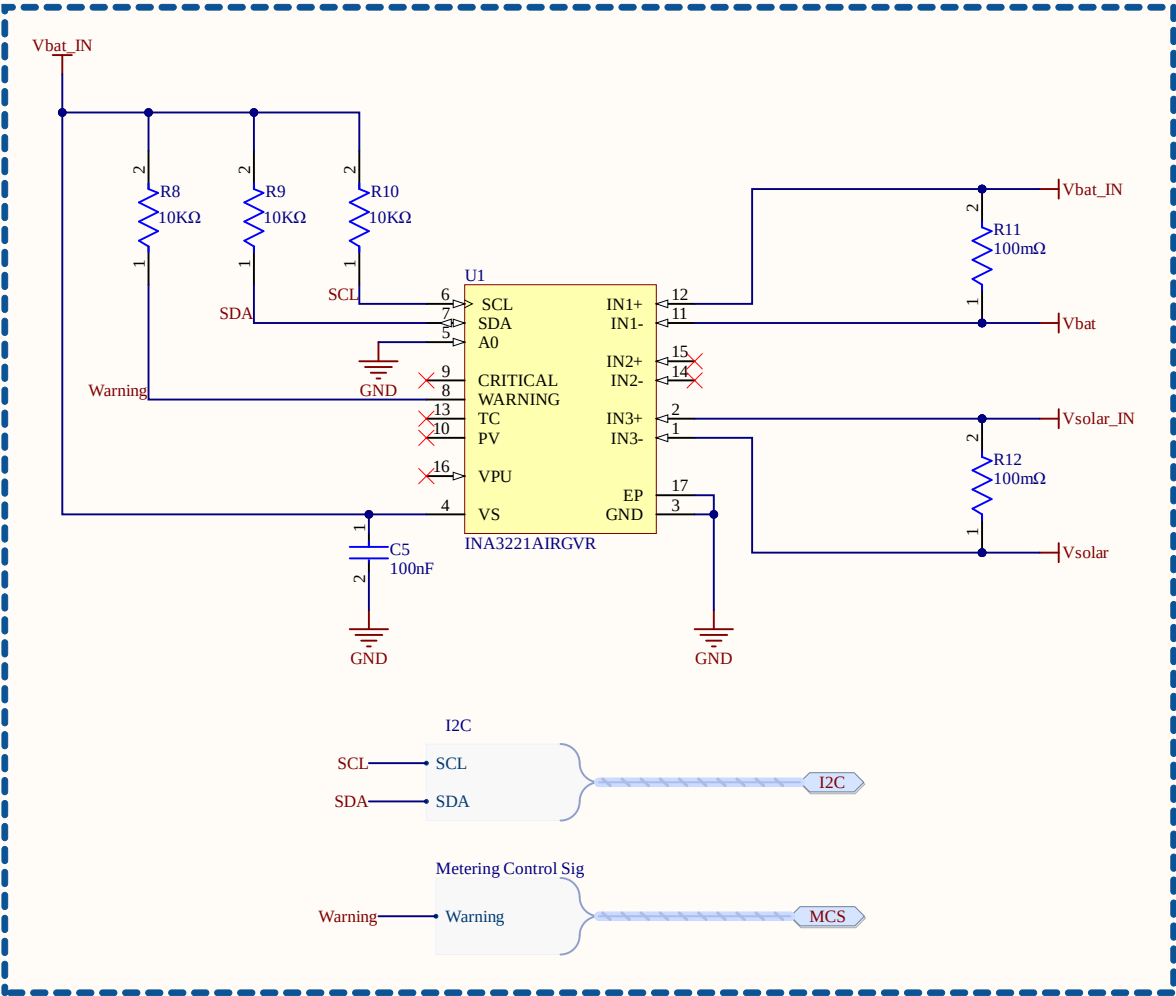
Short Circuit Protection
Short CKT Protection



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Block Diagram.SchDoc		

DC Power Metering Circuit

The INA3221 **U1** is a triple-channel, high-side current and bus voltage monitor with an I²C compatible interface. It is measuring the current, voltage and power provided by the solar panel and battery to the circuit and using 2 control pins i.e. "warning" and "TC". When the warning pin is low then it means the current or voltage is exciding the programmed limit. and the TC will be low when the conversion is complete



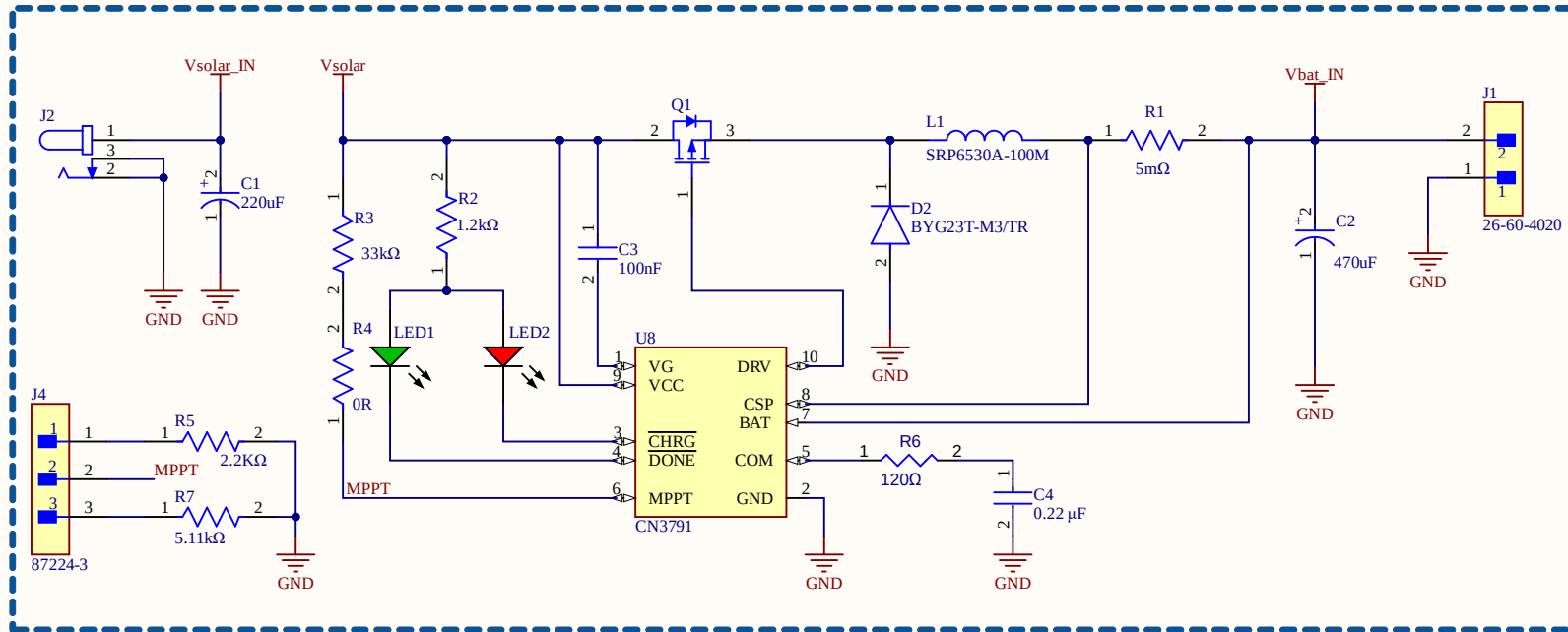
Pinout Description (16-Pin VQFN Package)

- Pins 12 & 11 (IN+1, IN-1): Shunt voltage inputs for Channel 1.
- Pins 15 & 14 (IN+2, IN-2): Shunt voltage inputs for Channel 2.
- Pins 1 & 2 (IN+3, IN-3): Shunt voltage inputs for Channel 3.
- Pin 4 (VS): Power supply (2.7 V to 5.5 V).
- Pin 3 (GND): Ground.
- Pin 16 (VPU): Pull-up voltage for Power-Valid output.
- Pin 10 (PV): Power-Valid alert output (open-drain).
- Pin 9 (Critical): Critical alert output (open-drain)
- Pin 8 (Warning): Warning alert output (open-drain).
- Pin 13 (TC): Timing Control alert output (open-drain).
- Pin 7 (SDA): I²C data line.
- Pin 6 (SCL): I²C clock line.
- Pin 5 (A0): I²C address selection.
- Thermal Pad: Solder to PCB for thermal stability.

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Battery Charging Circuit

The CN3791 [U8](#) is a standalone lithium-ion battery charger IC designed to efficiently charge single-cell lithium-ion batteries using a PWM switch-mode controller. It features a photovoltaic cell maximum power point tracking (MPPT) function, making it ideal for solar-powered applications. The IC operates in constant current and constant voltage modes, with a fixed regulation voltage of 4.2V ($\pm 1\%$ accuracy) and a programmable charge current up to 4A set by an external sense resistor. It supports a wide input voltage range of 4.5V to 28V and includes features like trickle charging for deeply discharged batteries, automatic recharge, charge termination, sleep mode, under-voltage lockout, and battery over-voltage protection.



Pin 1 (VG): Internal voltage regulator output that powers the gate driver, stabilized with a 100nF capacitor to VCC.

Pin 2 (GND): Ground pin, acting as the reference point for the IC's circuits and input supply.

Pin 3 (CHRG): Open-drain output that goes low during charging to indicate status, typically driving an LED.

Pin 4 (DONE): Open-drain output that signals charge completion by going low, otherwise high impedance.

Pin 5 (COM): Loop compensation input, stabilized with a 220nF capacitor and 120Ω resistor to ground.

Pin 6 (MPPT): Input for setting the photovoltaic cell's maximum power point, regulated to 1.205V via a resistor divider.

Pin 7 (BAT): Negative input for sensing charge current, connected to the battery via a sense resistor.

Pin 8 (CSP): Positive input for charge current sensing, paired with BAT to monitor current flow.

Pin 9 (VCC): Power supply input (4.5V to 28V), powering the IC with bypass capacitors for stability.

Pin 10 (DRV): Gate drive output that controls an external P-channel MOSFET for charging regulation.

$$V_{\text{MPPT}} = 1.205 \times \left(1 + \frac{R_2}{R_4}\right)$$

Where : $V_{MPPT} = 9V$

$$I_{CH} = \frac{0.120}{R_1}$$

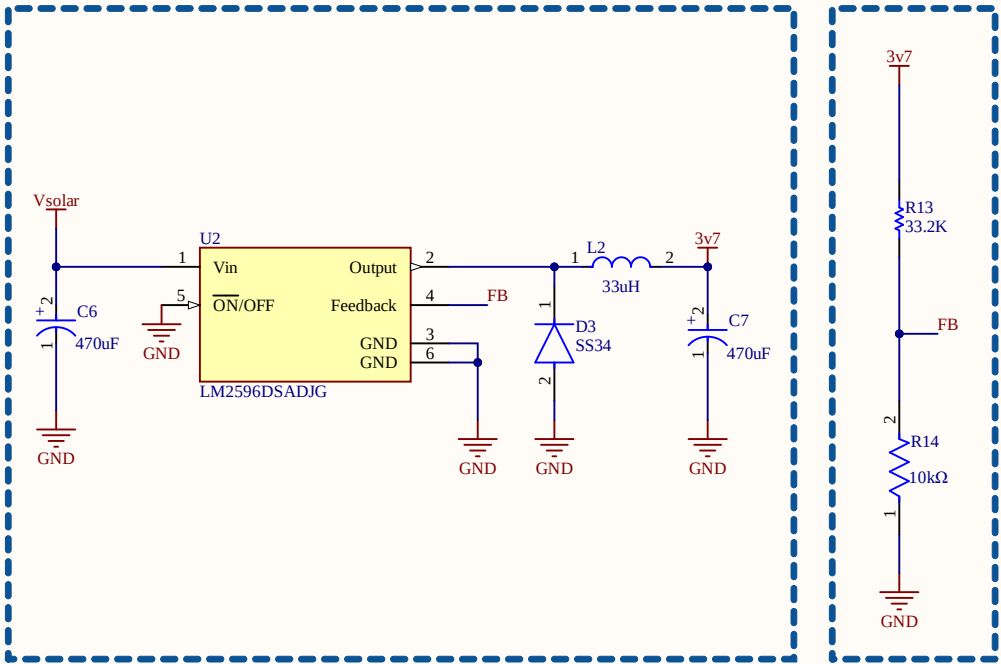
Where : $I_{CH} = 2.5A$

The solar panel should be used of rating 9V @ 20w (Voltage is set using R3 and R7) and the charging current of batter is set to 2.5A using resistor R1

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4V2 Buck Convertor

The LM2596 is a popular step-down (buck) switching regulator IC, with the adjustable version (LM2596-ADJ) capable of converting input voltages from 4.5V to 40V into a variable output voltage ranging from 1.2V to 37V, delivering up to 3A of continuous current. It operates at a fixed 150 kHz switching frequency, offering high efficiency and built-in features like thermal shutdown and current limiting. The IC requires minimal external components, making it ideal for compact power supply designs.



The undervoltage protection circuit is turn on only when the voltage from the solar pannel is greater than 4.1v
The BJT will remain off untill the Vsolar will cross the Zener voltage which turn on the BJT providing GND to the pin 5 of LM2595 **U2**

Operating Conditions:

Input Voltage (Vin): 4.5V to 40V, applied to pin 1 with a filter capacitor to reduce noise.

Output Voltage (Vout): Adjustable from 1.2V to 37V, set via resistors connected to the feedback pin, output through pin 2 with an LC filter.

ON/OFF Pin (Pin 5): Enables the IC when grounded or below 1.3V; pulling above 1.3V (up to 25V max) activates shutdown mode, reducing input current to ~80µA. Can be left open or grounded if shutdown isn't needed.

Feedback Pin (Pin 4): Senses output voltage to regulate it by adjusting the internal switch's duty cycle. External resistors set the desired output voltage ($V = 1.23v$).

Operating Current: Up to 3A continuous, with peak currents up to 6.9A.

Temperature Range: Typically -40°C to +125°C, with thermal shutdown protection.

Vbat = 4.1v, when the solar energy is available.

Vbat = battery's voltage when Solar energy is not sufficient

Title		
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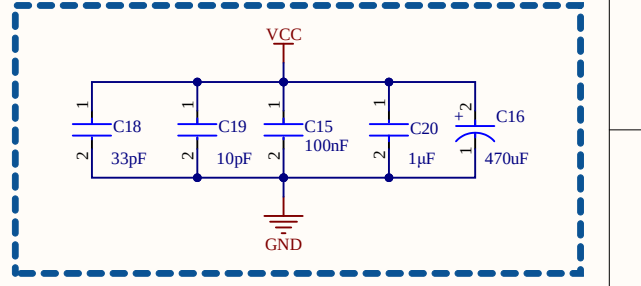
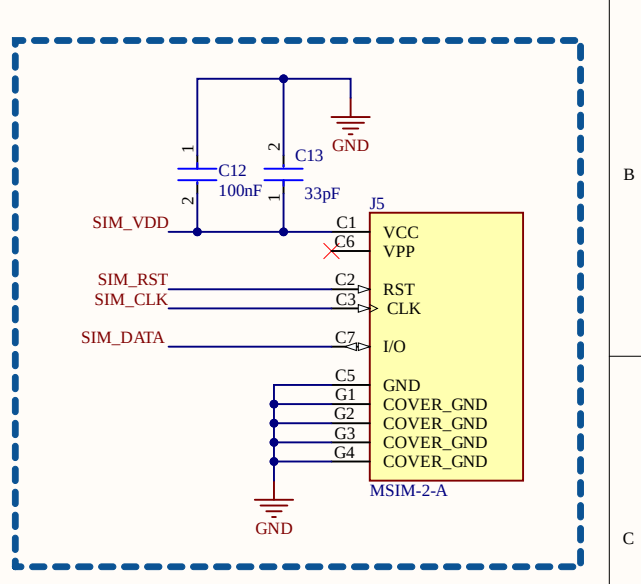
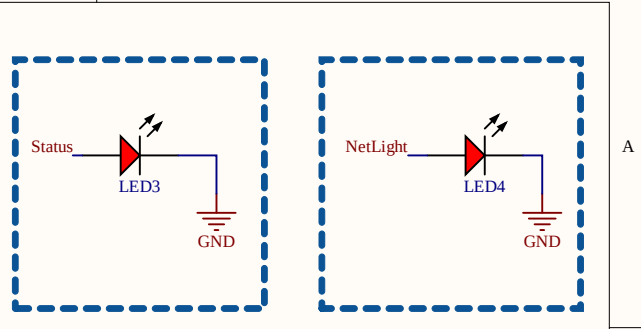
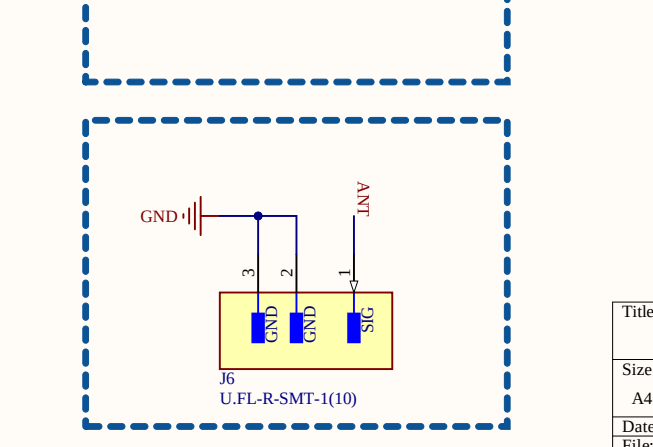
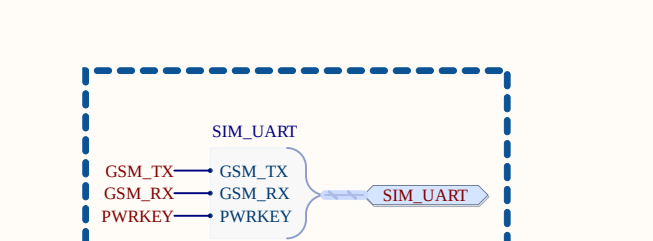
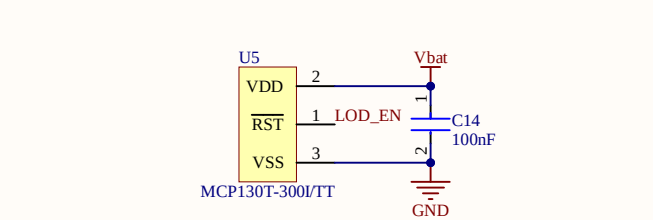
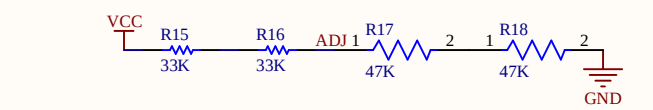
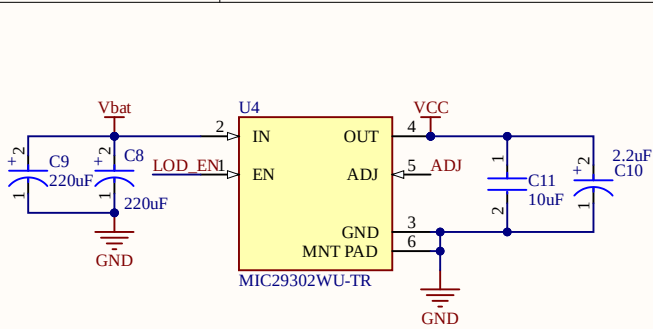
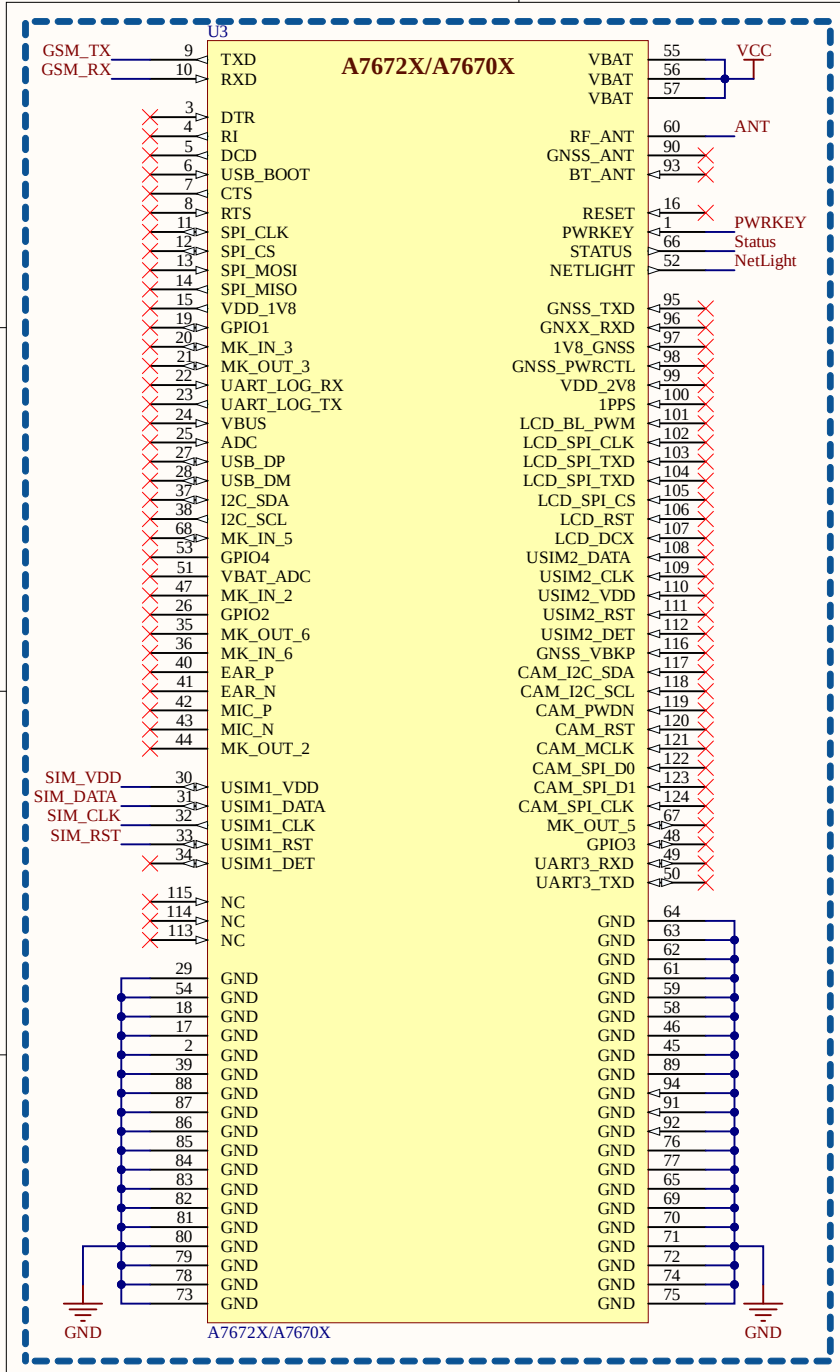
4G Sim Module

A

B

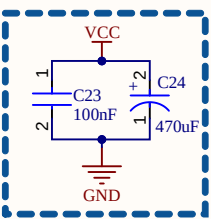
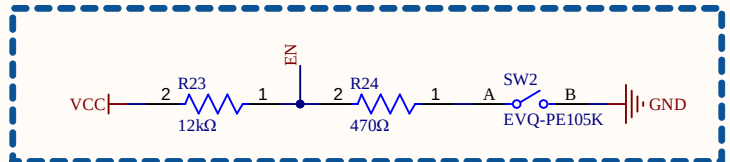
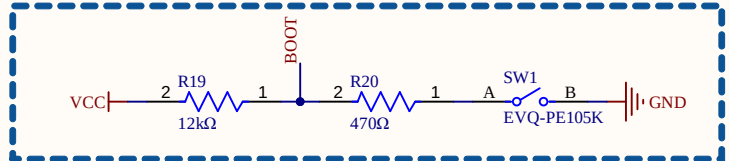
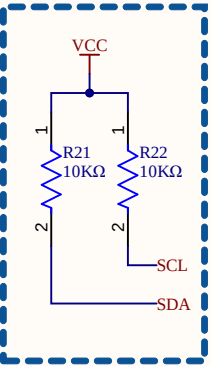
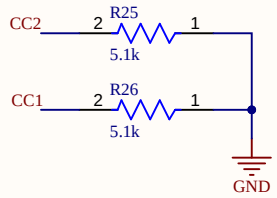
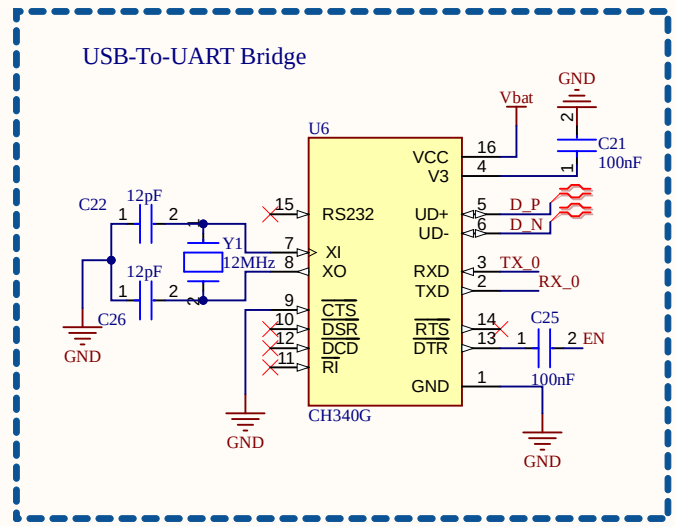
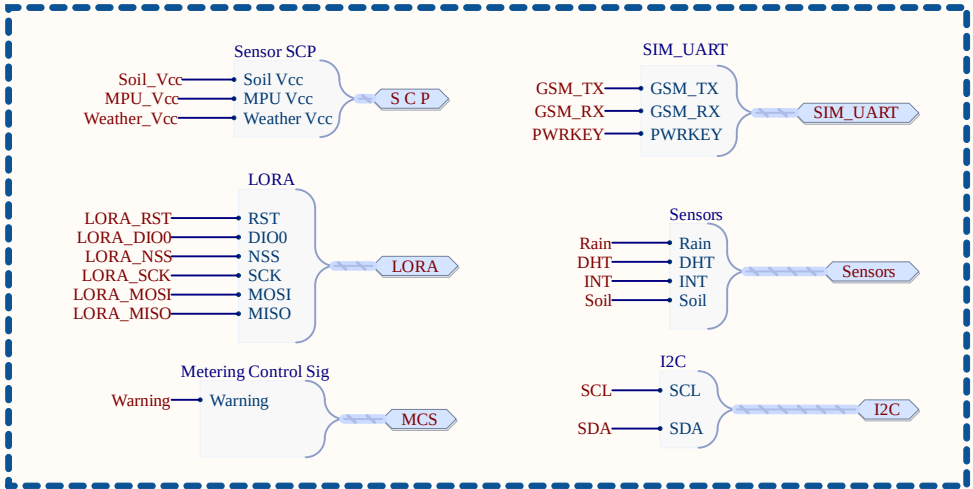
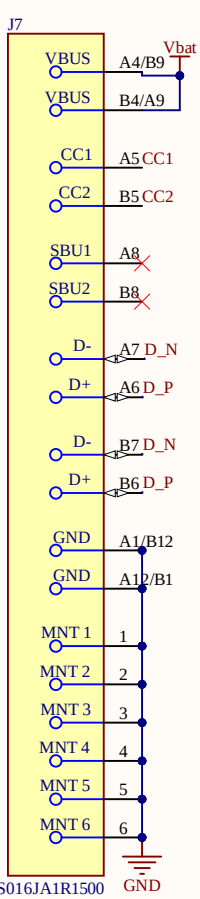
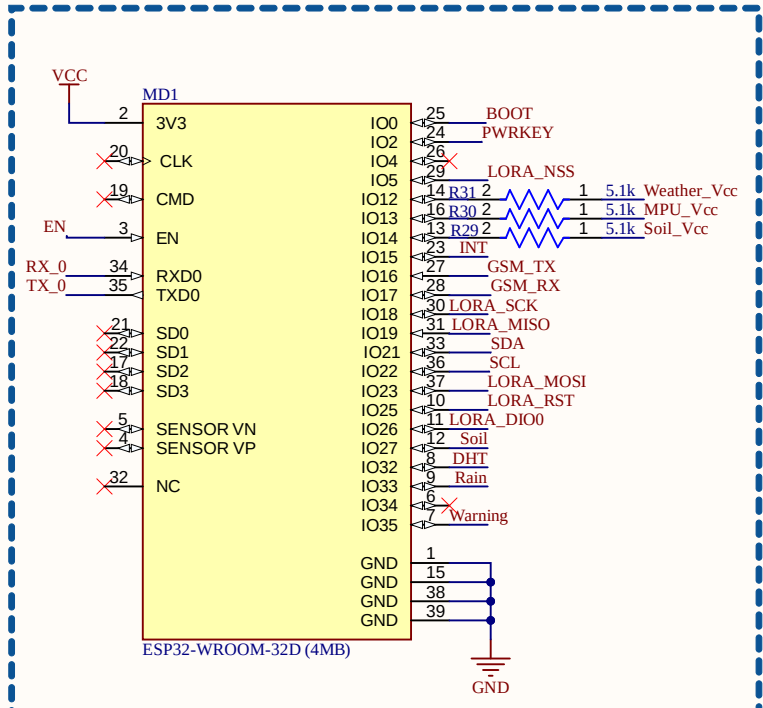
C

D



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File:	Sim A7670C.SchDoc	Drawn By:

Microcontroller

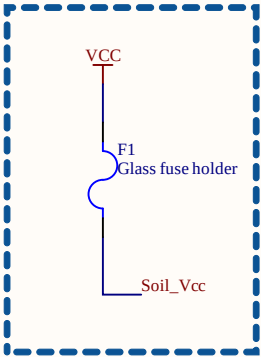


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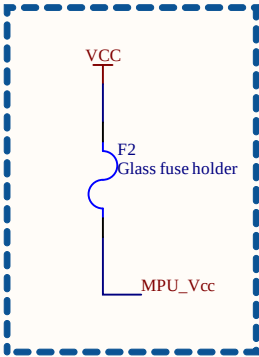
Short Circuit Protection

For short circuit protection 3 Fuses are used for weather node, MPU 6050 and soil moisture sensor, each uses a 100mA fues. In case of short circuit fuse will break the connection to stop the power suupply to the sensors. The sensor requies 3.3v to work, so the supply is continously moitored by the MCU by digital reading the pin

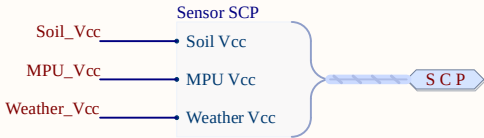
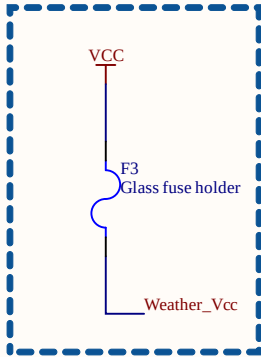
Soil Moisture Sensor



MPU6050 Sensor



Weather Node



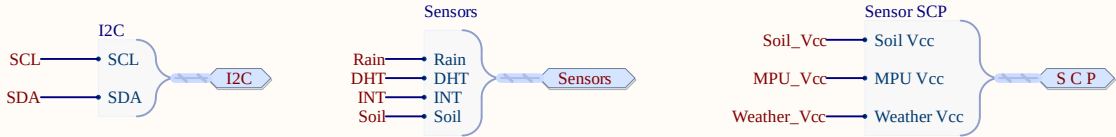
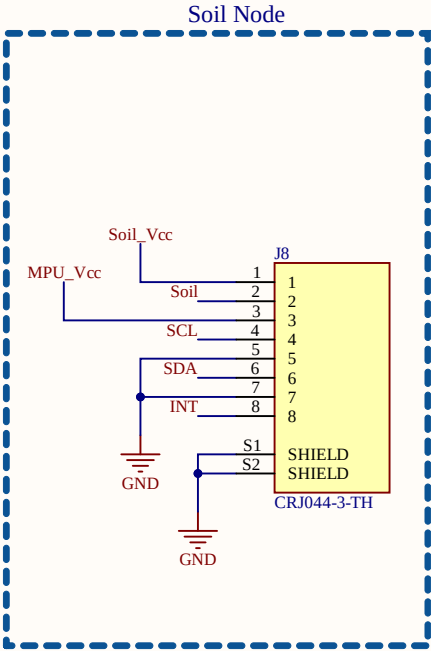
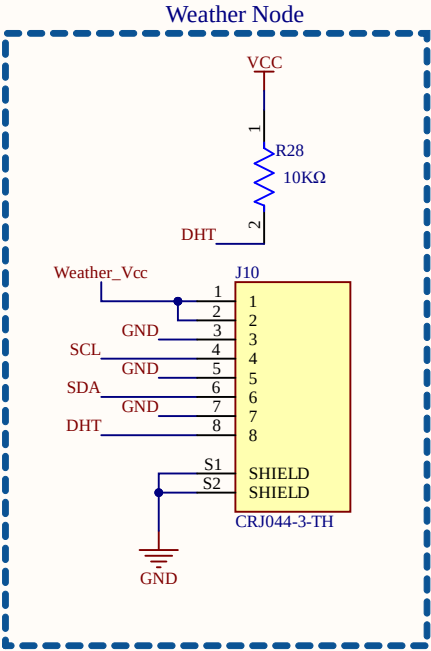
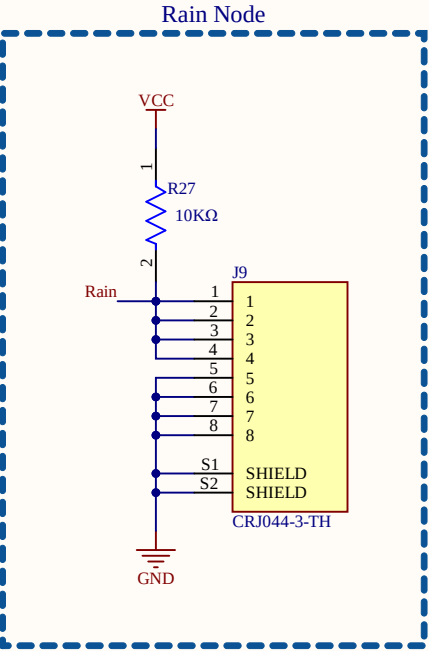
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Sensors

The Rain node is hall effect switch, which is in pulled up configuration when ever the switch changes the positom the reading is taken by the MCU GPIO.

Weather node contains a DHT22(tempraturea and moisture), BH1750(Luminous intensity) and BMP180(Pressure).
DHT -> Data pin of DHT22
I2C -> for BH1750 and BMP180
Weaterh_Vcc -> is the short ckt protection for the weather node

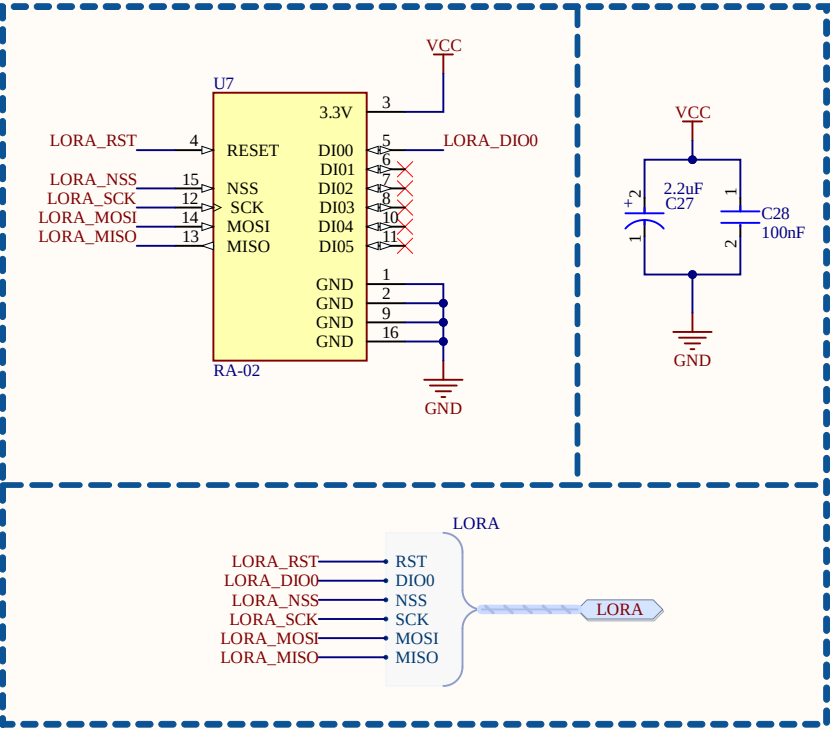
Soil Node contains Capacitive soil moisture sensors(Moisture of soil) and MPU6050(movement in land)
Soil moisture sensor -> Analog value of soil moistuer
MUP6050 -> Acclerometer and Gyroscope I2C pins
INT-> is the interrupt pin of the MPU6050
Soil_Vcc and MPU_Vcc is the short circuit protection circuit for the moisture sensor and MPU



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LORA RA-02

The LoRa RA-02 is a wireless communication module developed by Ai-Thinker, utilizing the Semtech SX1278 transceiver for long-range, low-power IoT applications. It employs LoRa spread spectrum technology, enabling communication distances up to 10 km with high sensitivity (-148 dBm) and strong anti-jamming capabilities. Operating in the 410–525 MHz frequency band (typically 433 MHz). The module uses an SPI interface for MCU connectivity and features air wake-up for low power consumption.



Frequency Range: 410–525 MHz (default 433 MHz)

Operating Voltage: 1.8–3.7 V (default 3.3 V)

Current Consumption: Receive <10.8 mA (LnaBoost closed, Band 1), Transmit <120 mA (+20 dBm), Sleep 0.2 μ A

Transmit Power: +18 $\pm$ 1 dBm (max +20 dBm)

Programmable Bit Rate: Up to 300 kbps (FSK), 62.5 kbps (LoRa)

Operating Temperature: -10°C to +85°C

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